

DATASHEET

EETX433A

Multifunctional RF Transmitter

Low-power RF solution for wireless remote control

TRANSMIT POWER

13 dBm

MODULATION

OOK / ASK

SHUTDOWN CURRENT

800 nA

TRANSMIT CURRENT

8 mA

VERSION

v1.0

PACKAGE INFORMATION

SOP8

DEVELOPED BY

Engeletron Engenharia

• Features

Transmit power	13 dBm
Modulation	OOK / ASK
Shutdown current	800 nA
Transmit current	8 mA
Frequencies	433.92 MHz
Supply voltage	1.8 V – 3.6 V
Package	SOP-8
Certifications	FCC, CE, ANATEL

• Overview

The EETX433A integrates a 433 MHz PLL oscillator, an RF transmitter, and a microcontroller (8-bit MCU core) on a single chip, delivering a simpler design, more flexible encoding, lower cost, and a smaller footprint — fully compliant with FCC, CE and ANATEL standards.

A built-in key-detection auto-shutdown feature minimizes standby current. The device also integrates output overcurrent protection, on-chip over-temperature protection, and supply undervoltage protection.

Building a complete wireless transmission system with the EETX433A requires only 2 inductors, 3 capacitors, resistors, a battery, and buttons — a fully integrated solution for remote control applications.

• Applications

- Security access control systems
- Wireless alarms
- Fire detection systems
- Wireless remote controls
- Home automation
- Ultra-low-power IoT sensors

• Block Diagram & Typical Application

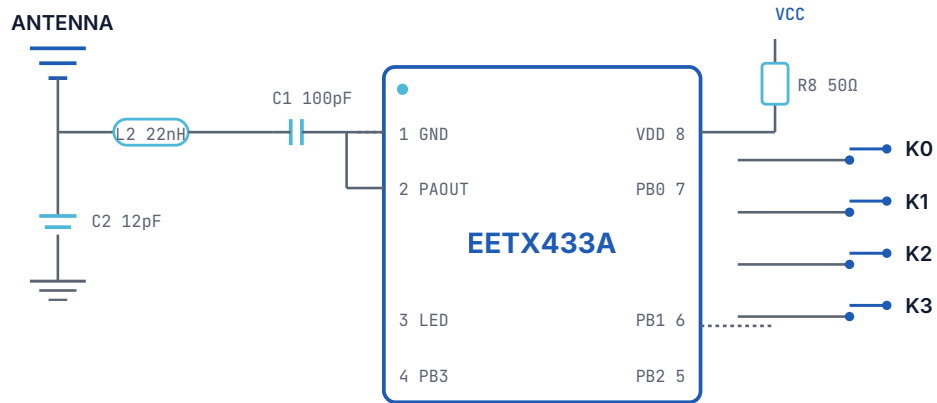


Figure 1: 433.92 MHz application diagram — 4 buttons

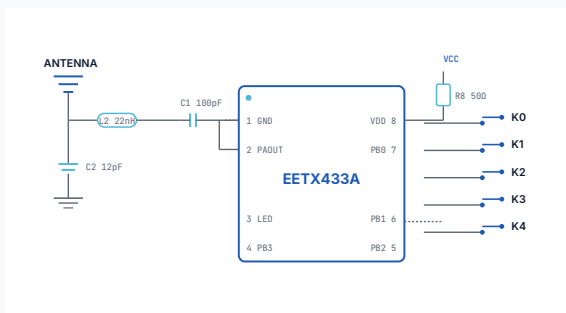


Figure 2: 433.92 MHz application diagram — 5 buttons

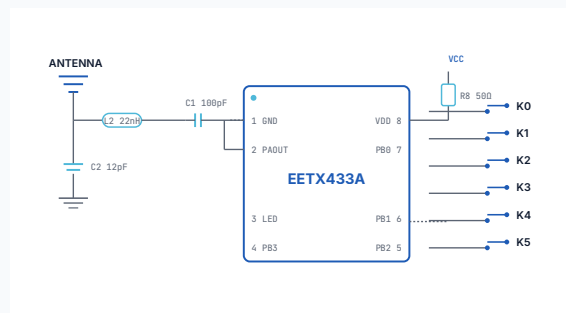


Figure 3: 433.92 MHz application diagram — 6 buttons

NOTES

- PIN 4 cannot be used as a common-key I/O.
- Fine-tuning C2 and C3 capacitance optimizes the antenna matching network, improving transmit power and efficiency.
- To ensure device reliability and lifetime, the absolute maximum ratings must not be exceeded. Sustained operation at these limits may cause permanent damage. In applications where input voltage may exceed VDD/GND, use an external protection diode.
- Combo MCU variant: additional inputs K0 and K1 are internally routed to PB4 and PB5, respectively.

Pinout & Pin Configuration



• Pin Description

Pin	Name	I/ O	Function
1	GND	I/ O	Power ground reference.
2	PAOUT	I	Power amplifier (PA) output.
3	LED	O	Data transmission indicator (LED driver).
4	PB3	I/ O	Bidirectional pin. Can be configured as input, open-drain output, or reset pin (RSTb — low level resets the MCU). Also serves as programming high-voltage input Vpp.
5	PB2	I/ O	Bidirectional pin. Timer external clock source (EX_CK1), PWM output, buzzer output (BZ1). Also acts as programming clock input (SCK).
6	PB1	I/ O	Bidirectional pin. When infrared mode is enabled, acts as IR carrier output. Also acts as programming data output (SDO).
7	PB0	I/ O	Bidirectional pin. When EIS=1 and INTIE=1, acts as external interrupt input. Also acts as programming data input (SDI).
8	VDD	I/ O	Positive supply voltage.

• Data Format

User-defined (configured via firmware in the embedded MCU).

• Electrical Characteristics

Absolute Maximum Ratings

Parameter	Symbol	Min.	Max.	Unit
VDD	V_DD	1.5	5.0	V
TA	T_A	-40	85	°C
TJ	T_J	-40	150	°C
TSTG	T_STG	-50	150	°C

Recommended Operating Conditions

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
VDD	V_DD		1.8	3	3.6	V
Ta	T_a		-30	25	85	°C
RL	R_L			50		Ω

DC Characteristics

$V_{SS}=0V$, $V_{DD}=3V$, $T_a=25^{\circ}C$

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
VDD	V_DD		2	3	3.6	V
IDD	I_DD	VDD=3V, 13dBm	6	8	10	mA
ISD	I_SD	D<0:3> floating		20	50	nA

• Analog / RF Characteristics

$V_{SS}=0V$, $V_{DD}=3V$, $T_a=25^{\circ}C$

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Po (433M)	P_0	RL=50Ω, VDD=3V		13		dBm
Po (433M)	P_0	RL=50Ω, VDD=3.6V		14		dBm
Dosc	D_osc	2V<VDD<3.6V			0.1%	kHz
Tf	T_f	VDD=3V, -30~70°C		10	16	ppm/°C
fr	f_r	VDD=3V, Ta=25°C			±100	kHz
fa	f_a	VDD=3V, Ta=25°C		±200	±250	kHz
tset	t_set			200		μs
P2	P_2	VDD=3V, Ta=25°C		-47		dB
P3	P_3	VDD=3V, Ta=25°C		-52		dB

• Soldering Guidelines

Because this component has low thermal capacity due to its compact construction, minimize exposure to external heat. Otherwise, thermal deformation may cause damage and characteristic shifts such as frequency drift. Use a non-corrosive rosin-type flux.

1. Hand soldering (test samples)

- Use a 30 W soldering iron with tip temperature 260–300 °C.
- Maximum contact time: 5 seconds.
- Keep the tip thoroughly clean.

2. Reflow soldering (SMT)

Follow the recommended temperature profile below: preheat up to 150 °C for no more than 60 seconds, then main heating up to 210 °C for no more than 10 seconds.

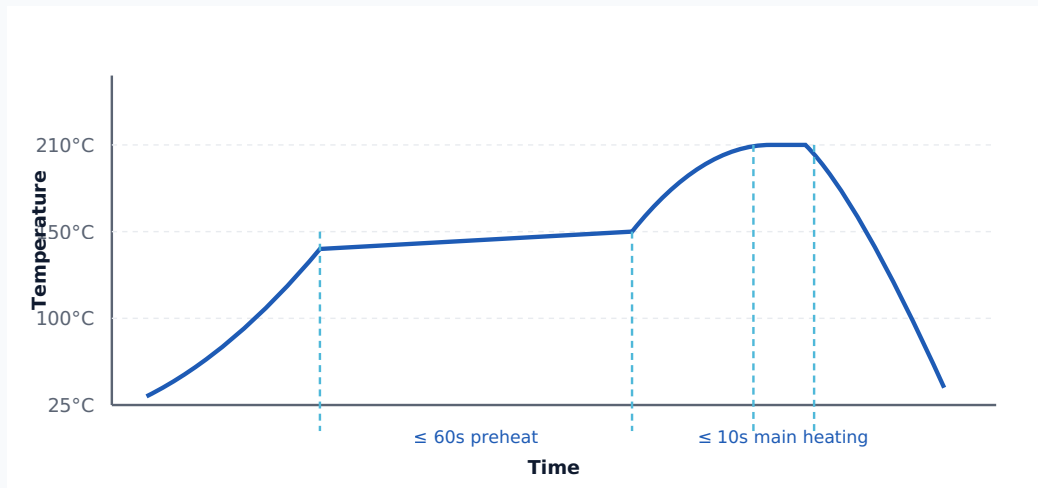


Figure 4: Recommended reflow temperature profile

• Package Information

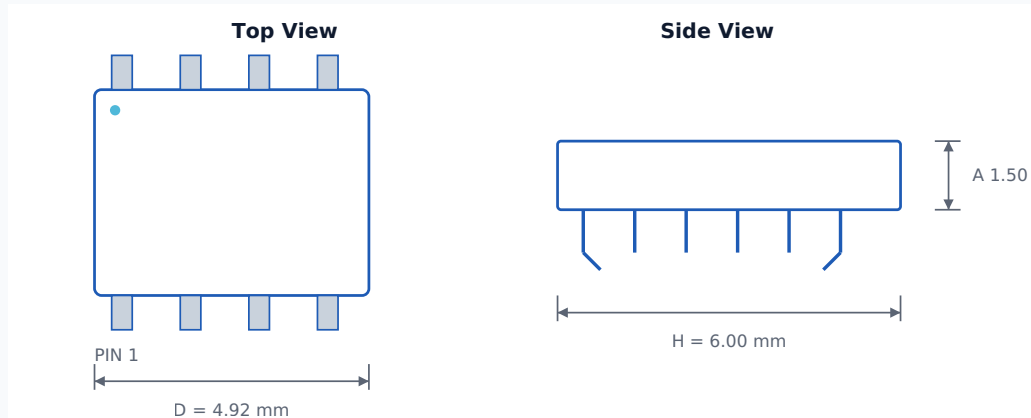


Figure 5: SOP-8 package — mechanical drawing

Symbol	Min (mm)	Nom (mm)	Max (mm)	Min (in)	Nom (in)	Max (in)
A	1.30	1.50	1.70	0.051	0.059	0.067
A1	0.06	0.16	0.26	0.002	0.006	0.010
b	0.30	0.40	0.55	0.012	0.016	0.022
c	0.15	0.25	0.35	0.006	0.010	0.014
D	4.72	4.92	5.12	0.186	0.194	0.202
E	3.75	3.95	4.15	0.148	0.156	0.163
e	—	1.27	—	—	0.050	—
H	5.70	6.00	6.30	0.224	0.236	0.248
L	0.45	0.65	0.85	0.018	0.026	0.033
θ	0°	—	8°	0°	—	8°

For more information about this product, visit: <https://engeletron.com.br/>

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